Laue Diffraction and X-Ray Crystallography

Parth Bhargava (A0310667E)
PC3193 Experimental Physics II

This two-part experiment uses Laue back-reflection diffraction and powder X-ray crystallography (XRC) to characterise crystalline materials. A Laue pattern recorded for a LiF single crystal shows a set of discrete, symmetrically arranged spots consistent with a well-ordered cubic crystal. Patterns for polyethylene monofilament demonstrate that mechanical drawing induces molecular alignment: the undrawn filament produces diffuse, isotropic scattering, while the drawn filament shows sharper, anisotropic features aligned along the draw axis. Powder XRC scans of two unknown polycrystalline samples are indexed using the $\sin^2\theta$ ratio method: Unknown 1 matches a diamond-cubic structure with $a = 5.430 \pm 0.002$ Å (silicon), and Unknown 2 matches a body-centred cubic structure with $a = 3.140 \pm 0.004$ Å (molybdenum). The 8-atom diamond-cubic structure factor is derived to explain the additional selection rules that distinguish diamond-cubic from simple FCC diffraction, specifically the extinction of reflections with $h + k + l \equiv 2 \pmod 4$.

1. INTRODUCTION

Determining the arrangement of atoms in a solid is one of the most direct ways to understand its physical properties. The electrical conductivity of silicon, the ductility of molybdenum, and the mechanical response of a polymer to stretching all follow from how their atoms or molecules are ordered. X-ray diffraction provides this structural information because X-ray wavelengths (roughly 0.5–2.5 Å) are comparable to interatomic spacings, so a crystal acts as a three-dimensional diffraction grating whose diffraction pattern directly reflects the underlying atomic arrangement.

Two complementary techniques exploit this. In Laue diffraction a polychromatic ("white") X-ray beam illuminates a single crystal. Because the beam contains a continuous range of wavelengths, each set of lattice planes (hkl) selects from the spectrum exactly the wavelength that satisfies the Bragg condition at the geometric angle set by the crystal orientation. The resulting pattern of spots on a photographic film reveals the point-group symmetry and orientation of the crystal. For partially ordered materials such as drawn polymers, the Laue pattern instead reveals the degree of molecular alignment through the anisotropy of the diffraction features.

In powder X-ray crystallography a monochromatic beam strikes a polycrystalline powder whose crystallites are randomly oriented. For every set of planes that satisfies the Bragg condition, a cone of diffracted radiation is produced; scanning the detector angle 2θ records intensity as a function of scattering angle, converting cones into discrete peaks. The positions of these peaks yield interplanar spacings, and their sequence identifies the crystal structure and lattice parameter. Together, the two methods span the range from qualitative symmetry assessment (Laue) to quantitative structure determination (powder XRC).

2. THEORY

2.1 X-Ray Production

X-rays are produced by two mechanisms, each yielding a different kind of spectrum suited to a different diffraction method.

When energetic electrons strike a target anode, they can eject inner-shell electrons from the target atoms. Outer-shell electrons then fill the vacancies, emitting X-rays at discrete energies characteristic of the electronic structure of the target element. For a copper anode, the dominant emission is Cu $K_{\alpha 1}$ at $\lambda = 1.5406$ Å. This effectively monochromatic radiation is essential for powder diffraction, where each sharp peak must correspond to a single well-defined wavelength.

The same accelerated electrons also lose energy when deflected in the Coulomb field of target nuclei, emitting bremsstrahlung, a continuous spectrum extending from visible wavelengths up to a cutoff set by the electron kinetic energy. This "white" beam is essential for Laue diffraction, where the continuous spectrum ensures that every set of lattice planes can find a wavelength satisfying the Bragg condition.

2.2 The Bragg Condition

Consider a set of parallel crystal planes separated by spacing d_{hkl}. An X-ray beam incident at glancing angle θ is partially reflected by each plane. Waves reflected from successive planes travel different path lengths; constructive interference occurs when the path difference equals an integer number of wavelengths:

$$n\lambda = 2d \sin\theta \tag{1}$$

This relation connects the measured angle 2θ (in a goniometer scan) to the interplanar spacing d, which contains the lattice geometry. In Laue diffraction the angle is fixed and the wavelength λ is the variable selected from the white beam; in powder diffraction λ is fixed (Cu $K_{\alpha 1}$) and θ is the scanned variable.

2.3 Laue Diffraction: Symmetry from a White Beam

In the Laue method, a crystal held at a fixed orientation is illuminated by the full bremsstrahlung spectrum. Each lattice plane (hkl) selects the wavelength $\lambda_{hkl} = 2d_{hkl}\sin\theta_{hkl}$ that satisfies Eq. (1) at its particular geometric angle. The resulting spots on the photographic film correspond to projections of reciprocal-lattice vectors, so the symmetry of the spot pattern directly reflects the point-group symmetry of the crystal as seen along the beam direction. For a cubic crystal viewed along a low-index zone axis, the pattern shows the characteristic rotational symmetry of that axis.

For polymer samples the Laue geometry is used differently. An amorphous or randomly oriented sample scatters X-rays diffusely. A mechanically drawn (stretched) sample, in which polymer chains are aligned along the draw axis, produces anisotropic scattering concentrated along preferred directions, giving a visual indicator of molecular orientation.

In the back-reflection arrangement used here, the film is placed between the X-ray source and the sample, so only diffracted beams with $2\theta > 90°$ reach the film. This restricts the visible spots to lattice planes nearly parallel to the sample surface, and means that only the shorter-wavelength portion of the bremsstrahlung spectrum contributes (since large θ requires small d or short λ to satisfy the Bragg condition). The back-reflection geometry is well suited to symmetry and orientation studies because the angular spread of spots is large, making symmetry elements easier to identify.

2.4 Powder Diffraction and Cubic Indexing

In a polycrystalline powder, crystallites are randomly oriented. For each family of planes satisfying the Bragg condition, a cone of diffracted radiation is produced. Scanning 2θ records intensity versus angle, converting these cones into peaks. For a cubic crystal with lattice parameter a, the interplanar spacing is:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \tag{2}$$

Combining with Eq. (1) (setting $n = 1$ and absorbing higher orders into the Miller indices):

$$\sin^2\theta = \frac{\lambda^2}{4a^2}\left(h^2 + k^2 + l^2\right) \tag{3}$$

This shows that for a cubic crystal, $\sin^2\theta$ increases in proportion to the integer $h^2 + k^2 + l^2$. By computing ratios of $\sin^2\theta$ values for successive peaks, the allowed sequence of $h^2 + k^2 + l^2$ is identified, which directly determines the crystal type.

2.5 Structure Factor and Selection Rules

Not all lattice planes produce observable reflections. The structure factor $F(hkl)$ determines the scattering amplitude from one unit cell:

$$F(hkl) = \sum_{j=1}^{M} f_j \exp\left[2\pi i\left(hx_j + ky_j + lz_j\right)\right] \tag{4}$$

where f_j is the atomic scattering factor of the j-th atom and (x_j, y_j, z_j) are its fractional coordinates. When $F(hkl) = 0$, destructive interference within the unit cell extinguishes the reflection entirely. These are the systematic absences.

For a **simple cubic** lattice (one atom at the origin), $F = f$ for all (hkl): no systematic absences.

For a **body-centred cubic** (BCC) lattice (atoms at $(0,0,0)$ and $(1/2,1/2,1/2)$):

$$F = f\left[1 + e^{i\pi(h+k+l)}\right] = \begin{cases} 2f & \text{if } h+k+l \text{ is even} \\ 0 & \text{if } h+k+l \text{ is odd} \end{cases} \tag{5}$$

For a **face-centred cubic** (FCC) lattice (atoms at $(0,0,0)$, $(1/2,1/2,0)$, $(1/2,0,1/2)$, $(0,1/2,1/2)$):

$$S_{\mathrm{FCC}} = f\left[1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}\right] = \begin{cases} 4f & \text{if } h,\ k,\ l \text{ all even or all odd} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

These selection rules generate distinct sequences of allowed $h^2 + k^2 + l^2$ values, which serve as fingerprints for crystal structure identification:

- **SC:** 1, 2, 3, 4, 5, 6, 8, 9, …
- **BCC:** 2, 4, 6, 8, 10, 12, …
- **FCC:** 3, 4, 8, 11, 12, 16, 19, 20, 24, …

The ratios of successive $\sin^2\theta$ values directly distinguish the three cubic types.

2.6 Diamond-Cubic Structure Factor

The diamond-cubic structure (silicon, germanium, and diamond) consists of two interpenetrating FCC sublattices displaced by $(1/4,1/4,1/4)$. The 8 atoms in the conventional unit cell occupy:

Sublattice 1: $(0,0,0)$, $(1/2,1/2,0)$, $(1/2,0,1/2)$, $(0,1/2,1/2)$
Sublattice 2: $(1/4,1/4,1/4)$, $(3/4,3/4,1/4)$, $(3/4,1/4,3/4)$, $(1/4,3/4,3/4)$

Since sublattice 2 is just sublattice 1 shifted by $(1/4,1/4,1/4)$, the total structure factor factorises as:

$$F(hkl) = S_{\mathrm{FCC}} \cdot \left[1 + e^{i\pi(h+k+l)/2}\right] \tag{7}$$

where S_{FCC} is the FCC structure factor from Eq. (6). The second factor $P = 1 + e^{i\pi(h+k+l)/2}$ introduces additional extinctions beyond the basic FCC rule:

TABLE I. Structure factor magnitude for the diamond-cubic 8-atom basis. Only FCC-allowed reflections are listed. Reflections with $h + k + l \equiv 2 \pmod 4$ are extinguished by destructive interference between the two sublattices.

$h+k+l \bmod 4$	$e^{i\pi(h+k+l)/2}$	P	$\lvert F\rvert^2$
0	1	2	$64f^2$
1	i	$1+i$	$32f^2$
2	-1	0	0
3	$-i$	$1-i$	$32f^2$

The physical consequence is that reflections such as (200) ($h + k + l = 2$) and (222) ($h + k + l = 6$) are forbidden in diamond-cubic but allowed in simple FCC. This additional extinction is

the experimental signature that distinguishes diamond-cubic from FCC.

3. PART A: LAUE DIFFRACTION

3.1 Aim

Determine the crystal quality and symmetry of a LiF single crystal from its Laue back-reflection pattern, and assess the degree of molecular orientation induced by mechanical drawing in polyethylene monofilament.

3.2 Methods

All experiments were performed on enclosed X-ray systems with radiation shielding and interlock mechanisms. X-rays were generated only when the scatter shield was locked in place, and tube current was monitored throughout.

LiF single crystal. A LiF mini-crystal was mounted with adhesive tape over the 1 mm diameter primary beam collimator (582.002), with the longitudinal axis aligned at approximately 45° to the beam. The auxiliary slide carriage was fitted with the collimator to the lead glass dome. X-ray film (38 mm × 35 mm) was loaded into the film cassette (562.013) without a primary beam shield, and the pattern was exposed for 40 minutes at experimental station E.S. 3.

Polyethylene monofilament. A polyethylene monofilament (585.006) was mounted vertically, centred over the 1 mm collimator port. A first exposure (7 minutes at E.S. 3, 30 kV, 75 μA) recorded the pattern for the as-received filament. The central section was then manually stretched until it acquired a clearly drawn, oriented form. A second exposure was recorded with the drawn portion centred over the beam, keeping all other parameters identical so that differences in the patterns reflect orientation changes rather than geometric drift.

3.3 Results

The LiF pattern (Fig. 1a) shows approximately 20–25 discrete, well-separated spots distributed about the central direct-beam position, with no diffuse background. The pattern has approximate mirror symmetry about at least one axis.

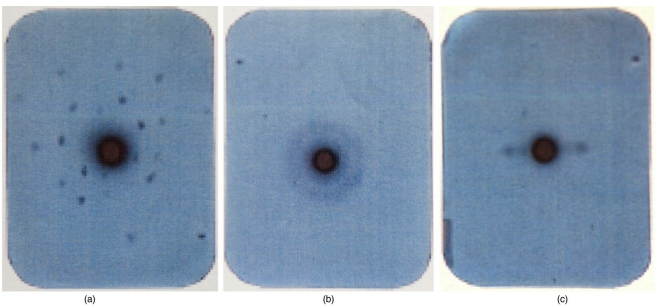


FIG. 1. Laue back-reflection patterns. (a) LiF single crystal: discrete spots arranged symmetrically about the direct beam. (b) Undrawn polyethylene: diffuse, isotropic scattering. (c) Drawn polyethylene: sharper features concentrated along the draw axis.

The undrawn polyethylene (Fig. 1b) produces a largely diffuse pattern with no sharp spots. The scattering is approximately isotropic; no preferred direction is apparent. After drawing (Fig. 1c), the pattern develops sharper features aligned along a preferred axis, with the scattering noticeably concentrated in one direction.

3.4 Analysis

LiF single crystal. The sharpness and isolation of the LiF spots confirm a well-ordered single crystal with long-range periodicity, consistent with cubic symmetry. LiF crystallises in the rock-salt (NaCl-type) structure with space group $Fm\bar{3}m$, belonging to the cubic point group $m\bar{3}m$, which possesses two-, three-, and four-fold rotation axes. Since the crystal was mounted with its axis at approximately 45° to the beam, not coinciding with any principal zone axis ($\langle 100\rangle$, $\langle 110\rangle$, or $\langle 111\rangle$), the pattern is a general projection of the reciprocal lattice rather than one showing the full rotational symmetry of a specific zone. The overall spot arrangement is still consistent with cubic symmetry.

Polyethylene. The contrast between the undrawn and drawn polyethylene patterns shows that drawing induces molecular orientation. In the undrawn state, any crystalline lamellae are randomly oriented, producing no preferred diffraction directions. Drawing forces the polymer chains into approximate alignment along the draw axis, creating a fibre texture: the c-axis of the orthorhombic polyethylene unit cell ($a = 7.40$ Å, $b = 4.93$ Å, $c = 2.534$ Å, space group $Pnam$) becomes preferentially parallel to the fibre direction. In reciprocal space this alignment smears the discrete reciprocal-lattice points into arcs centred on the fibre axis, producing the streaked, anisotropic pattern observed in Fig. 1c rather than the sharp spots of a true single crystal. The drawn filament is not a single crystal (the scattering is still much more diffuse than the LiF pattern), but it is no longer isotropic.

3.5 Discussion

The key physical insight from Part A is that long-range periodicity and short-range order produce qualitatively different diffraction signatures under the same experimental geometry. The LiF crystal, with its periodic lattice extending over macroscopic distances, satisfies the Bragg condition at sharply defined angles for each (hkl) plane, giving discrete spots. The polyethylene, even after drawing, retains only partial crystalline order: the chains align preferentially along the draw axis but the crystallites remain small and their orientations are distributed around the fibre direction. This partial ordering is enough to break the isotropy of the scattering but not enough to produce the sharp spots that would indicate a single-crystal-like arrangement.

The back-reflection geometry limits the experiment to qualitative conclusions. Because only diffracted beams with $2\theta > 90°$ reach the film, the number of visible spots is small relative to the full reciprocal lattice, and measuring inter-spot angles precisely from a photograph is difficult. Quantitative lattice parameter determination or complete symmetry assignment from a Laue pattern requires either mounting along a known zone axis or using computational indexing tools (e.g. a Greninger chart), neither of which was attempted here. For the polyethylene, the film-based detection provides only a visual comparison between oriented and isotropic scattering; extracting a quantitative

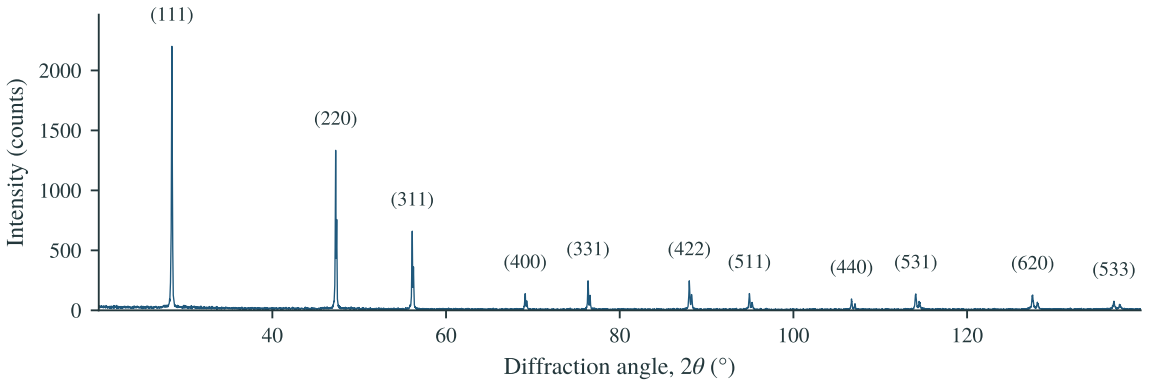


FIG. 2. Powder diffraction scan of Unknown 1 (Sample Run 11; Cu K_α, 0.02° steps), with each peak labelled by its Miller indices. The dominant peak at $2\theta \approx 28.4°$ is (111); every observed reflection has h, k, l all odd or all even with $h + k + l \neq 2 \pmod 4$, the diamond-cubic selection rule.

orientation distribution would require a digital area detector and azimuthal intensity analysis.

3.6 Improvements

- Mount the LiF crystal along a known zone axis to display the full rotational symmetry of the cubic point group.
- Use a Greninger chart or software indexing to assign (hkl) labels to individual spots and confirm the zone axis.
- Replace photographic film with a digital area detector (e.g. image plate) for quantitative intensity analysis, shorter exposure times, and the ability to compute azimuthal intensity profiles for fibre-textured samples.
- Record a Laue pattern of a non-cubic crystal (e.g. quartz) for comparison with the high-symmetry cubic case.

4. PART B: POWDER X-RAY CRYSTALLOGRAPHY

4.1 Aim

Identify two unknown polycrystalline samples by indexing their powder X-ray diffraction patterns using the $\sin^2\theta$ ratio method, determining the crystal structure type and lattice parameter of each, and comparing the results with literature values.

4.2 Methods

Two unknown polycrystalline samples were scanned on a PANalytical X'Pert^3 powder diffractometer in θ–2θ (goniometer) geometry using a Cu anode ($K_{\alpha1} = 1.5406$ Å, $K_{\alpha2} = 1.5444$ Å, intensity ratio $K_{\alpha2}/K_{\alpha1} = 0.50$). Scans covered $2\theta = 20°$–$140°$ with a step size of 0.02° and 0.40 s per step. The complete instrument settings for each sample are recorded in Table VIII. Diffraction data were exported as CSV files for peak identification and indexing. Peaks were identified by visual inspection of the plotted intensity-vs.-2θ scans: a feature was accepted as a Bragg peak if it rose clearly above the local background with a profile consistent with the instrumental resolution. The 2θ position of each peak was taken as the channel with the maximum count rate; for doublet-split peaks at high angles, the stronger $K_{\alpha1}$ component was used for indexing.

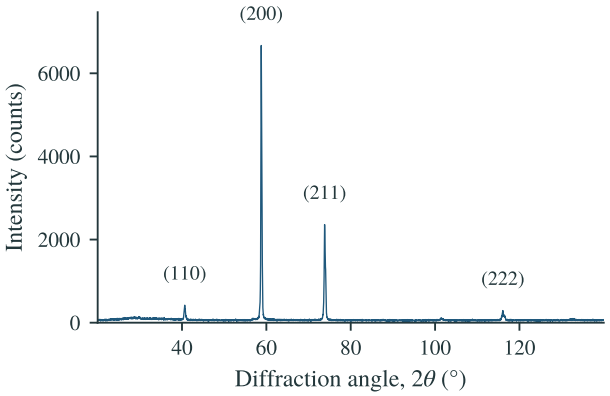


FIG. 3. Powder diffraction scan of Unknown 2 (Sample Run 7), labelled as in Fig. 2. Only reflections with $h + k + l$ even appear, as required for a body-centred cubic lattice; the dominant peak at $2\theta \approx 58.8°$ is (200). Weak features between 26° and 34° sit near the noise floor.

4.3 Results

The Unknown 1 scan (Fig. 2) shows 11 well-resolved peaks between $2\theta = 28°$ and 137°, with a dominant peak at 28.43° (2201 counts). Several high-angle peaks are visibly split into doublets. The Unknown 2 scan (Fig. 3) shows four strong peaks, the largest at 58.81° (6318 counts), plus a series of weak features between 26° and 34° with intensities of 110–140 counts.

4.4 Analysis

4.4.1 Unknown 1: $\sin^2\theta$ Ratio Analysis

The first step in powder pattern analysis is to identify the crystal type from the sequence of observed reflections. Computing $\sin^2\theta$ for each peak and forming ratios relative to the first strong peak reveals the allowed values of $h^2 + k^2 + l^2$:

TABLE II. Ratio analysis for Unknown 1. The sequence 3, 8, 11, 16, 19, 24 matches FCC-allowed values, but the absence of $h^2 + k^2 + l^2 = 4$ (the (200) reflection) is the signature of diamond-cubic structure.

2θ (°)	$\sin^2\theta$	**Ratio**	$h^2 + k^2 + l^2$	(hkl)
28.43	0.0606	1.00	3	(1 1 1)
47.33	0.1614	2.66	8	(2 2 0)
56.15	0.2222	3.67	11	(3 1 1)
69.11	0.3226	5.32	16	(4 0 0)
76.35	0.3833	6.32	19	(3 3 1)
88.01	0.4844	7.99	24	(4 2 2)

The ratio sequence $1 : 2.66 : 3.67 : 5.32 : 6.32 : 7.99$ matches $h^2 + k^2 + l^2 = 3, 8, 11, 16, 19, 24$. These are the values allowed by the FCC selection rule (all even or all odd indices), but with a conspicuous absence: $h^2 + k^2 + l^2 = 4$, corresponding to the (200) reflection, does not appear. Neither does $h^2 + k^2 + l^2 = 12$, corresponding to (222). These are precisely the reflections extinguished by the diamond-cubic structure factor (Table I): both have $h + k + l \equiv 2 \pmod 4$, where the phase factor $P = 0$ due to destructive interference between the two FCC sublattices. For a simple FCC metal (aluminium, copper, gold), the (200) reflection would be among the strongest peaks. Its complete absence here, together with the lattice parameter, confirms diamond-cubic structure.

4.4.2 Unknown 1: Lattice Parameter and Identification

TABLE III. Full indexed peak list for Unknown 1. Every observed reflection satisfies $h + k + l \not\equiv 2 \pmod 4$, confirming the diamond-cubic structure factor. The lattice parameter a clusters tightly around 5.430 Å.

(hkl)	2θ (°)	**Intensity**	d (Å)	a (Å)	$h + k + l$
(1 1 1)	28.43	2201	3.137	5.433	3
(2 2 0)	47.33	525	1.919	5.428	4
(3 1 1)	56.15	256	1.637	5.427	5
(4 0 0)	69.11	139	1.358	5.432	4
(3 3 1)	76.35	246	1.246	5.430	7
(4 2 2)	88.01	246	1.109	5.431	8
(5 1 1)	94.93	140	1.045	5.430	7
(4 4 0)	106.69	95	0.960	5.430	8
(5 3 1)	114.07	129	0.918	5.431	9
(6 2 0)	127.53	122	0.859	5.431	8
(5 3 3)	136.89	71	0.828	5.430	11

The lattice parameters computed from all 11 peaks cluster tightly around $a = 5.430$ Å with a sample standard deviation of 0.002 Å (range 5.427–5.433 Å), which matches the literature value for silicon ($a_{\text{lit}} = 5.431$ Å) very well. The last column of Table III checks the diamond-cubic selection rule: every observed reflection has $h + k + l \not\equiv 2 \pmod 4$, as predicted by Eq. (7) and Table I. Unknown 1 is therefore identified as **silicon**.

4.4.3 Unknown 2: $\sin^2\theta$ Ratio Analysis

TABLE IV. Ratio analysis for Unknown 2. The sequence 2, 4, 6, 12 is a subset of BCC-allowed values ($h + k + l$ even), confirming body-centred cubic structure.

2θ (°)	$\sin^2\theta$	**Ratio**	$h^2 + k^2 + l^2$	(hkl)
40.67	0.1208	1.00	2	(1 1 0)
58.81	0.2409	1.99	4	(2 0 0)
73.83	0.3606	2.98	6	(2 1 1)
116.07	0.7190	5.95	12	(2 2 2)

The ratios $1 : 2.00 : 2.98 : 5.95$ match $h^2 + k^2 + l^2 = 2, 4, 6, 12$, which is the BCC selection rule: only reflections with $h + k + l$ even are allowed. Peaks with $h^2 + k^2 + l^2 = 8$ and 10, corresponding to (220) and (310), are expected at $2\theta \approx 88°$ and $\approx 101°$ but are absent. Since all BCC reflections share the same structure-factor magnitude $|F|^2 = 4f^2$, these are not selection-rule extinctions. Two factors account for their absence. First, the Lorentz-polarisation factor $L_p = \big(1 + \cos^2 2\theta\big)/\big(\sin^2\theta \cos\theta\big)$ drops by roughly a factor of 5 between $2\theta = 41°$ (110) and $2\theta = 88°$ (220), and the Debye–Waller thermal factor e^{-2M} and atomic scattering factor $f(\sin\theta/\lambda)$ both decrease monotonically with angle, compounding the suppression. Second, the observed intensity pattern already departs from ideal powder statistics: (110) registers only ≈ 350 counts despite having the largest $p \times L_p$ product of any BCC reflection ($p = 12$, $L_p \approx 14$), suggesting preferred orientation (texture) that redistributes intensity away from certain reflections. With the glass-holder scattering producing a noise floor of 110–140 counts between 26° and 34°, even modest additional suppression at higher angles places (220) and (310) below the detection limit.

The weak features between $2\theta \approx 26°$ and 34° do not index consistently with the BCC phase and are attributed to amorphous scattering from the glass sample holder, which produces a broad halo centred near $2\theta \approx 22°$–$28°$ for Cu K_α radiation.

4.4.4 Unknown 2: Lattice Parameter, Zero-Offset, and Identification

The lattice parameters computed from the four indexed peaks show a systematic trend: 3.135, 3.138, 3.142, and 3.145 Å for (110), (200), (211), and (222) respectively, monotonically increasing with 2θ. This is the classic signature of a small positive zero-angle offset $\Delta(2\theta)$ in the goniometer. A positive offset shifts all measured 2θ values upward, depressing d (and hence a) more strongly at low angles where $\partial(\sin\theta)/\partial\theta$ is smaller. The correction is described by:

$$a_{\text{obs}} \approx a_{\text{true}}\left(1 - \frac{\Delta(2\theta)}{2\tan\theta}\right)$$

The trend is qualitatively consistent with $\Delta(2\theta) \approx +0.04°$. A Nelson–Riley extrapolation addresses this by plotting a against the function $\text{NR} = 1/2\big(\cos^2\theta/\sin\theta + \cos^2\theta/\theta\big)$ (with θ in radians) and extrapolating to $\text{NR} = 0$:

TABLE V. Nelson–Riley extrapolation data for Unknown 2. A linear fit to a vs. NR gives an intercept $a(0) = 3.146$ Å, in close agreement with $a_{\text{lit}} = 3.147$ Å for molybdenum (0.03% discrepancy).

(hkl)	2θ (°)	NR	a (Å)
(1 1 0)	40.67	2.50	3.135
(2 0 0)	58.81	1.51	3.138
(2 1 1)	73.83	1.03	3.142
(2 2 2)	116.07	0.30	3.145

The linear fit yields $a(0) = 3.146$ Å with a slope of -0.0046 Å per unit NR, reducing the discrepancy from 0.2% (uncorrected mean) to 0.03%. With only four peaks the fit is not strongly constrained, but the monotonic trend and the improved agreement with the literature value support the zero-offset interpretation.

The mean lattice parameter is $a = 3.140$ Å with a sample standard deviation of 0.004 Å (range 3.135–3.145 Å), identifying the material as **molybdenum**.

TABLE VI. Summary of lattice parameters and structural identification for both unknowns.

Sample	Mean a (Å)	Structure	Identification
Unknown 1	5.430 ± 0.002	Diamond cubic	Silicon
Unknown 2	3.140 ± 0.004	BCC	Molybdenum

4.4.5 $K_{\alpha1}/K_{\alpha2}$ Doublet Splitting

At high 2θ the Cu $K_{\alpha1}$ and $K_{\alpha2}$ lines ($\Delta\lambda = 0.0038$ Å) are resolved as separate peaks. Differentiating Bragg's law, the angular separation scales as $\Delta(2\theta) \propto 1/(\sin\theta\cos\theta)$, which diverges as $\theta \to 90°$. The Si data, with 11 peaks spanning a wide angular range, provide a clean test of this. Five clearly resolved doublet pairs are observed:

TABLE VII. Observed and calculated $K_{\alpha1}/K_{\alpha2}$ doublet separations for Si. The monotonic increase with angle and the close agreement with theory confirm that these are instrumentally resolved spectral components, not artefacts.

(hkl)	$2\theta_{K\alpha1}$ (°)	$2\theta_{K\alpha2}$ (°)	$\Delta(2\theta)_{\text{obs}}$ (°)	$\Delta(2\theta)_{\text{calc}}$ (°)
(3 3 1)	76.35	76.57	0.22	0.22
(4 2 2)	88.01	88.27	0.26	0.27
(5 1 1)	94.93	95.25	0.32	0.32
(5 3 1)	114.07	114.51	0.44	0.47
(6 2 0)	127.53	128.11	0.58	0.64

$\Delta(2\theta)$ increases monotonically with angle, as expected from the Bragg-law derivative. The small discrepancies at the highest angles ($\approx 0.06°$ at $2\theta = 128°$) reflect the difficulty of locating the $K_{\alpha2}$ peak when it overlaps with the tail of the stronger $K_{\alpha1}$ component. The $K_{\alpha2}$ intensities are approximately half those of $K_{\alpha1}$, consistent with the literature ratio of 0.50.

4.5 Discussion

Peak intensities and the Lorentz-polarisation factor. For diamond-cubic Si, the strongest reflection is (111) ($I \approx 2200$ counts), followed by (220) (≈ 525) and (311) (≈ 256). The (111) peak dominates despite the structure factor relation $|F_{111}|^2 = 32f^2 < |F_{220}|^2 = 64f^2$ because the Lorentz-polarisation factor $L_p = \big(1 + \cos^2 2\theta\big)/\big(\sin^2\theta\cos\theta\big)$ is roughly 7 times larger at $2\theta = 28.4°$ than at 47.3°, more than compensating for the smaller $|F|^2$.

For BCC Mo, the (200) reflection dominates (≈ 6300 counts) despite all BCC reflections sharing the same $|F|^2 = 4f^2$. This is because (200) at $2\theta = 58.8°$ has a favourable combination of multiplicity ($p = 6$) and L_p value, while higher-angle peaks are penalised by the Debye–Waller factor. The relatively weak (110) peak (≈ 350 counts) at the lowest angle may reflect a degree of preferred orientation (texture) in the powder sample.

Peak widths and crystallite size. The observed full-width at half-maximum (FWHM) of the Si and Mo peaks is on the order of 0.1°–0.15° in 2θ, comparable to the instrumental resolution set by the divergence slit, receiving slit, and Soller slit geometry. This indicates that crystallite sizes in both samples are well above the regime where Scherrer broadening ($\beta \approx K\lambda/(L\cos\theta)$, with $K \approx 0.9$) would produce measurable excess width, corresponding to crystallites larger than roughly 100 nm. No meaningful crystallite-size information can therefore be extracted from these scans.

Systematic uncertainties. The main source of uncertainty in the XRC analysis is peak-position determination, which depends on the step size (0.02°), counting statistics, and any systematic zero-angle offset. The Si data show a tight clustering of lattice parameters (5.427–5.433 Å, $\sigma = 0.002$ Å) because the 11 peaks span a wide 2θ range (28°–137°), so random position errors partially cancel in the mean and any systematic offset averages out. The Mo data, with only four peaks spanning a narrower range, show a clearer systematic trend that reveals the goniometer zero-offset ($\approx +0.04°$), which depresses the low-angle lattice parameter estimates.

4.6 Improvements

- Use an internal standard (e.g. a known Si powder mixed with the unknown) to calibrate and correct the goniometer zero-offset in situ, rather than relying on post-hoc extrapolation.
- Fit peak profiles with Voigt or pseudo-Voigt functions rather than reading peak maxima by eye, to improve position accuracy (especially for overlapping doublets) and to extract peak widths for Scherrer analysis.
- Extend the scan to higher 2θ to capture additional Mo peaks, improving the statistical basis for lattice parameter determination and the Nelson–Riley extrapolation.
- Use Rietveld refinement to simultaneously fit all structural and instrumental parameters, providing a more rigorous identification and better uncertainty estimates.

5. CONCLUSION

The two parts of this experiment illustrate complementary uses of X-ray diffraction. Laue diffraction with a white beam tests whether a sample possesses long-range crystalline order and, if so, what symmetry that order carries; powder diffraction with a monochromatic beam goes further, extracting the lattice parameter and crystal structure type from the precise positions of Bragg peaks. The Laue patterns showed that the LiF crystal is a well-ordered cubic single crystal and that mechanical drawing of polyethylene induces molecular alignment detectable as anisotropic scattering, demonstrating that diffraction is sensitive

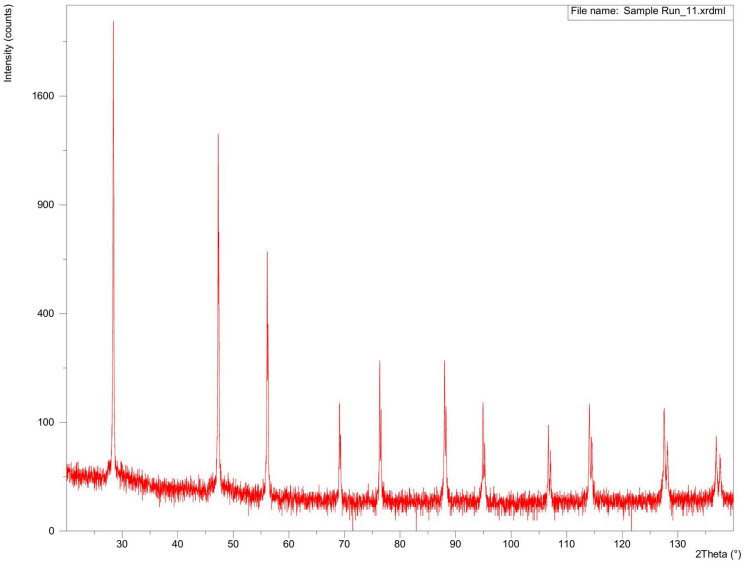


FIG. 4. Raw PANalytical Data Viewer output for Unknown 1 (Sample Run 11). The dominant peak near $2\theta \approx 28°$ and the series of progressively weaker peaks are visible without annotation.

not only to the presence of periodicity but also to its degree and directionality.

The powder XRC analysis identified Unknown 1 as diamond-cubic silicon ($a = 5.430 \pm 0.002$ Å) and Unknown 2 as BCC molybdenum ($a = 3.140 \pm 0.004$ Å). The silicon identification relied on the additional selection rule introduced by the two-sublattice diamond-cubic structure: reflections with $h + k + l \equiv 2 \pmod 4$ are extinguished, a pattern that would not occur for a simple FCC metal. The molybdenum identification was strengthened by a Nelson–Riley extrapolation that corrected a small goniometer zero-offset, improving the lattice parameter agreement with the literature value from 0.2% to 0.03%. These results show how systematic absences, peak ratios, and systematic-error analysis work together to turn a set of diffraction peaks into a structural identification.

APPENDIX

Instrument Settings

TABLE VIII. XRC instrument settings for the two unknown samples, extracted from the measurement metadata.

Parameter	**Unknown 1 (Run 11)**	**Unknown 2 (Run 7)**
Diffractometer	PANalytical X'Pert^3	PANalytical X'Pert^3
Goniometer	PW3050/60 (θ/θ)	PW3050/60 (θ/θ)
Geometry	Flat sample stage (PW3071/xx)	Flat sample stage (PW3071/xx)
Anode material	Cu	Cu
$K_{\alpha1}$ wavelength	1.5406 Å	1.5406 Å
$K_{\alpha2}$ wavelength	1.5444 Å	1.5444 Å
Generator voltage	40 kV	45 kV
Tube current	30 mA	40 mA
Scan range (2θ)	20.0°–140.0°	20.0°–140.0°
Step size	0.020°	0.020°
Scan type	Continuous	Continuous
Time per step	0.40 s	0.40 s
Divergence slit	Fixed, 1.52 mm	Fixed, 1.52 mm
Anti-scatter slit	Fixed 1° (1.52 mm)	Fixed 2° (3.03 mm)
Receiving slit	0.10 mm	0.10 mm
Soller slits	0.04 rad (inc. + diff.)	0.04 rad (inc. + diff.)
K_β filter	Ni, 0.020 mm	Ni, 0.020 mm
Detector	Proportional Xe (point)	Proportional Xe (point)
No. of points	6000	6000
Scan date	15 Feb 2024	8 Oct 2025

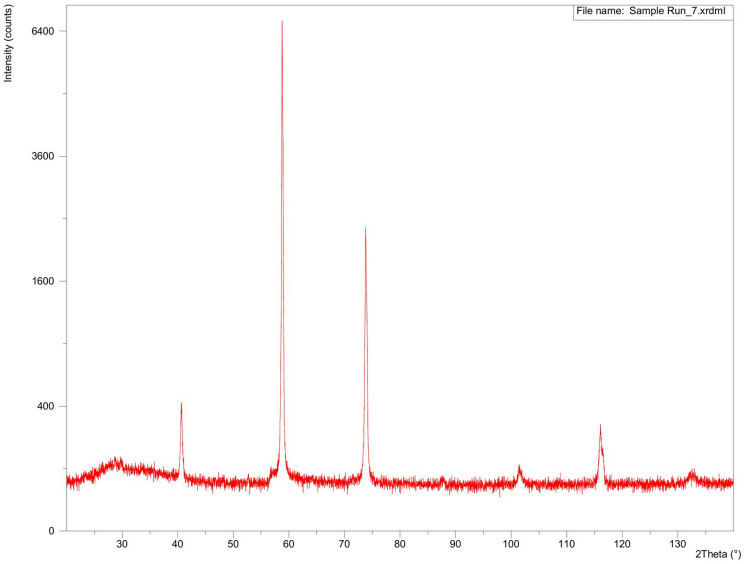


FIG. 5. Raw PANalytical Data Viewer output for Unknown 2 (Sample Run 7). The strong (200) peak at $2\theta \approx 59°$ dominates; the broad hump between 26° and 34° from amorphous scattering is visible near the baseline.

Laue Exposure Log

TABLE IX. Exposure parameters for Laue diffraction patterns.

Sample	Station	Exposure time	Tube settings
LiF mini-crystal	E.S. 3	40 min	Standard (1 mm collimator, 45° alignment)
Polyethylene (undrawn)	E.S. 3	7 min	30 kV, 75 μA
Polyethylene (drawn)	E.S. 3	7 min	30 kV, 75 μA

Raw Diffractometer Output

The following screenshots from the PANalytical Data Viewer show the unprocessed scans as recorded by the instrument, prior to peak annotation. They serve as a check that the annotated plots in the main text faithfully represent the measured data.

REFERENCES

[1] Laue Diffraction, (n.d.).
[2] X-Ray Crystallography, (n.d.).
[3] C. Kittel, *Introduction to Solid State Physics*, 8th ed. (Wiley, 2005).
[4] H. P. Myers, *Introductory Solid State Physics* (Taylor & Francis, 1997).

DECLARATION ON THE USE OF GENERATIVE AI

I declare that I **HAVE** used generative AI tools to produce this assignment.

I acknowledge that generative AI was used in the following manner:

TABLE X. AI Tool Usage Declaration

AI Tool Used	My Prompt and AI Output	How the Output Was Used
Claude Code	**Prompt:** "Help me process the powder XRD data files, identify peaks, calculate $\sin^2\theta$ ratios, and determine lattice parameters for the unknown samples." **Output:** Python script for peak extraction and lattice parameter calculation	Used for data processing automation. All diffraction measurements were recorded by me, and the identification of crystal structures through systematic absence analysis and lattice parameter comparison is my own work.
Claude Code	**Prompt:** "Help with Typst formatting for figures, tables, equations, and appendix layout. Check that cross-references and file paths compile correctly." **Output:** Typst formatting guidance and layout suggestions	Used only for formatting. All experiment content, analysis, and scientific interpretations were written and verified by me.